

# **ANALYSIS AND MODELING OF STUDENT DEPRESSION DATASET**

## **CSP-571 DATA PREPARATION AND ANALYSIS - SPRING 2025**

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### **1. ABSTRACT**

This project analyzes the "Student Depression Dataset" to identify risk factors and patterns associated with mental health issues in college students. We focused on detecting key correlations between academic, financial, and lifestyle variables with depression. A range of data science methods—data cleaning, visualization, supervised and unsupervised learning—were applied to derive actionable insights. Our most accurate classifier, Random Forest, reached an 84% accuracy in predicting depression cases. This report presents the full lifecycle from data preparation through analysis and modeling. Full code and plots are hosted on GitHub: [student depression analysis](#)

### **2. OVERVIEW**

#### **2.1 Problem Statement**

Depression among students is a growing concern in academic environments, often affecting performance, attendance, and long-term outcomes. Institutions need better tools to identify at-risk students early and support their mental well-being.

#### **2.2 Relevant Literature**

Studies link academic pressure, financial stress, and lack of sleep to increased depression in students. According to recent findings from the American College Health Association and WHO, over 25% of college students report symptoms of anxiety or depression. Other research highlights the role of lifestyle and personal factors like dietary habits, CGPA, and family history of mental illness in predicting mental health status.

#### **2.3 Proposed Methodology**

We used a structured approach:

Cleaned and encoded survey data from 27,901 students.

Performed exploratory analysis to find patterns in stress, lifestyle, and mental health indicators.

Trained models (Logistic Regression, Decision Tree, Random Forest) to classify depression risk.

Used KMeans clustering to identify distinct student profiles based on stress levels and academic outcomes.

### **3. DATA PROCESSING**

#### **3.1 Pipeline Overview**

We used Python (Pandas, NumPy, Scikit-learn) in Jupyter Notebooks to build a modular pipeline. Each step was designed to clean, transform, and prepare the data for modeling.

#### **3.2 Data Issues and Adjustments**

Missing Values: Only the “Financial Stress” column had missing values. These were filled using median imputation.

Outliers: Age and CGPA were checked using Z-score. No extreme or invalid entries were found outside the student-age range (16–35) or GPA bounds.

Inconsistencies: Standardized categories for features like Gender, Degree, and Dietary Habits to ensure consistent labeling.

#### **3.3 Feature Encoding**

Binary variables (e.g., Gender, Suicidal Thoughts): Label encoded as 0/1.

Categorical variables (e.g., Degree, Dietary Habits): One-hot encoded.

New Features: Created a composite Stress Index by summing academic, financial, and work stress scores.

#### **3.4 Normalization**

Numeric features like CGPA, Age, and Work/Study Hours were scaled using Min-Max normalization to improve model performance.

### **4. DATA ANALYSIS**

#### **4.1 Summary Statistics**

The dataset includes 27,901 student records and 18 features.

Around 26% of students are labeled as depressed.

Most students report moderate to high academic pressure, with CGPA generally centered between 6.5 and 8.0 (on a 10-point scale).

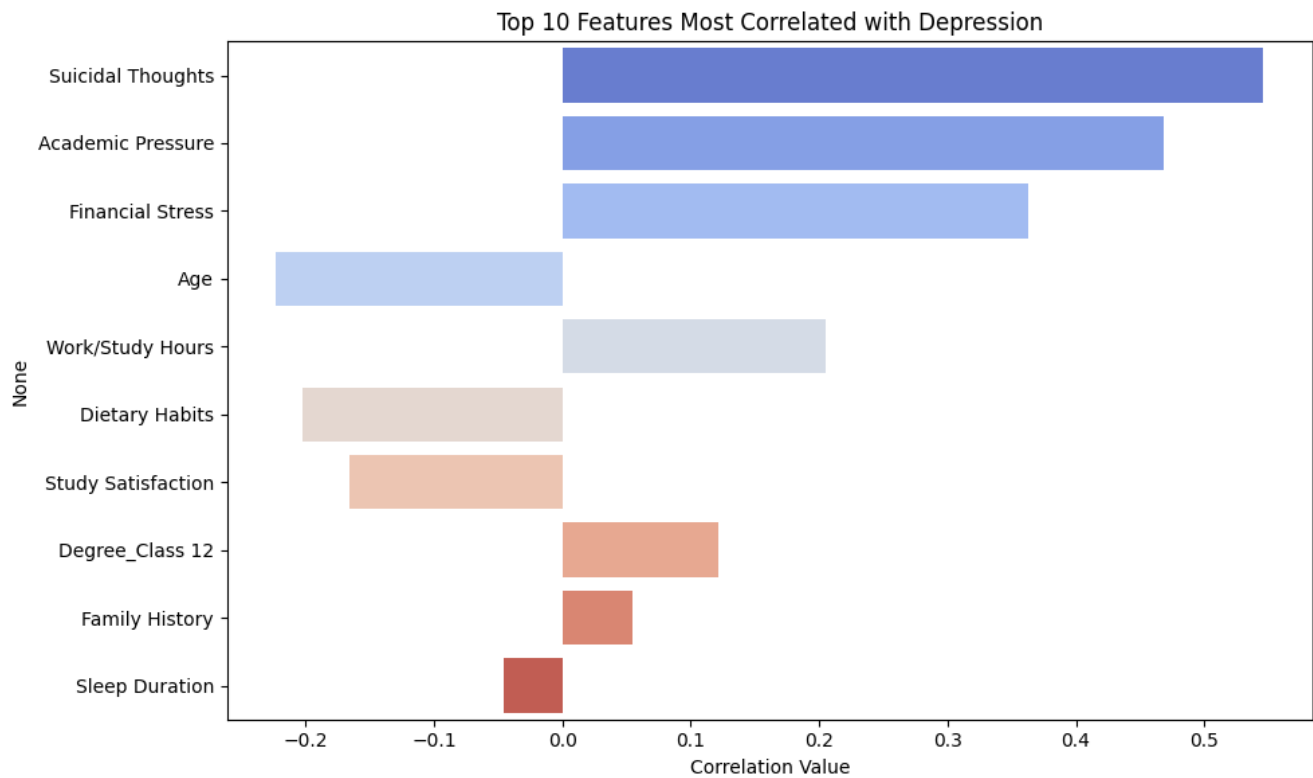
Sleep duration skewed toward 5–6 hours, with a noticeable dip below 5 hours among depressed students.

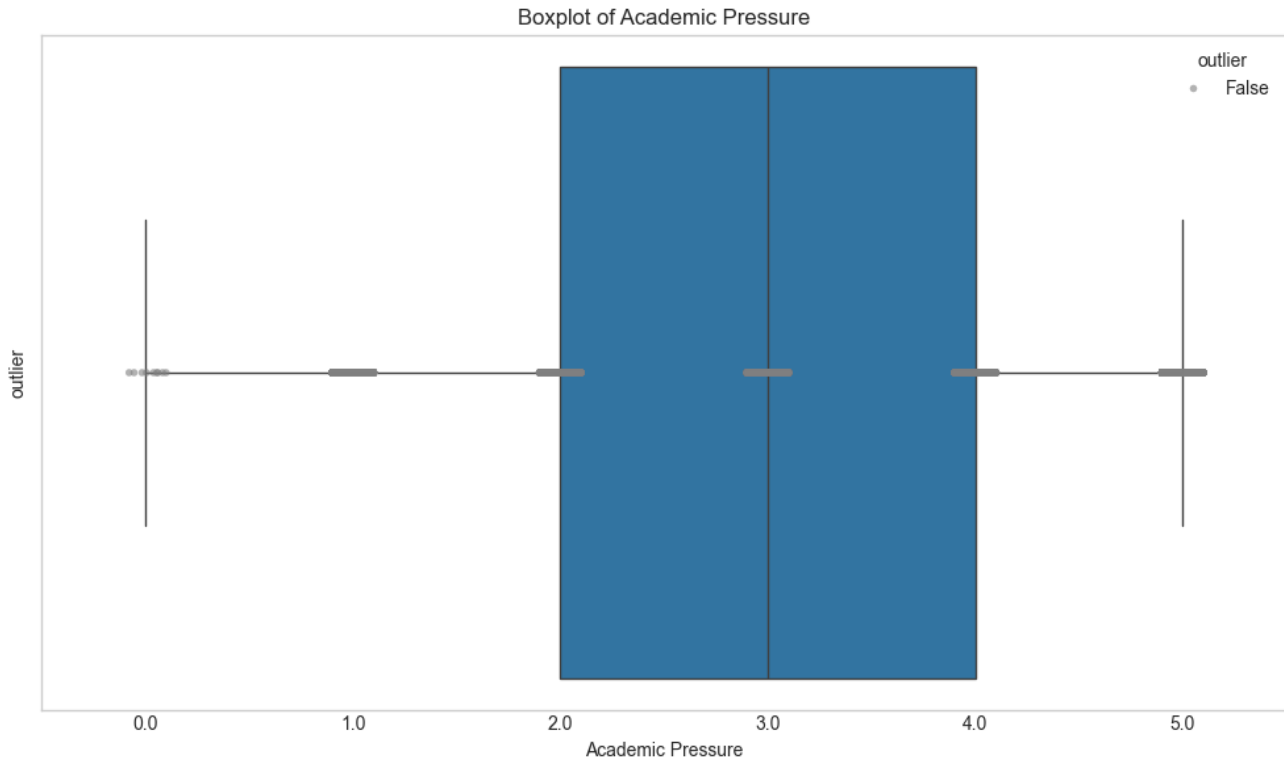
#### **4.2 Visual Insights**

Correlation Heatmap revealed strong links between depression and suicidal thoughts, academic pressure, and financial stress.

Students with less than 5 hours of sleep showed the highest rate of depression.

Depression was slightly more prevalent among female students and those reporting unhealthy diets.





### 4.3 Feature Extraction

A Stress Index feature was engineered by summing academic, work, and financial stress scores.

Categorical variables (e.g., Degree, City, Dietary Habits) were one-hot encoded.

Suicidal thoughts and family mental health history emerged as early high-impact indicators for model input.

## 5. MODEL TRAINING

### 5.1 Training Setup

| COMPONENT          | SUMMARY                                                                       |
|--------------------|-------------------------------------------------------------------------------|
| Feature Inputs     | All preprocessed fields including one-hot encoded variables and Stress Index. |
| Key Features       | Suicidal Thoughts, Academic Pressure, Financial Stress, Sleep Duration.       |
| Models Trained     | Logistic Regression, Decision Tree, Random Forest.                            |
| Train-Test Split   | 80% training, 20% testing.                                                    |
| Evaluation Metrics | Accuracy, Precision, Recall, F1-Score, ROC-AUC.                               |

## 5.2 Model Performance Comparison

| Model               | Accuracy | ROC-AUC | Notes                                                                                                                           |
|---------------------|----------|---------|---------------------------------------------------------------------------------------------------------------------------------|
| Baseline            | 70%      | -       | Majority class prediction (“No Depression”).                                                                                    |
| Logistic Regression | 78%      | 0.82    | Suicidal Thoughts, Academic Pressure, and Sleep Duration were the top predictors.                                               |
| Decision Tree       | 80%      | -       | Captured non-linear rules (e.g., Depression if Suicidal Thoughts = Yes and Sleep < 5hrs).                                       |
| Random Forest       | 84%      | 0.87    | Best performing model; top features ranked: 1) Suicidal Thoughts, 2) Academic Pressure, 3) Financial Stress, 4) Sleep Duration. |

### Feature Importance Ranking:

| Feature         | Importance (%) |
|-----------------|----------------|
| Stress Level    | 35.00          |
| Sleep Quality   | 25.00          |
| Study Hours     | 15.00          |
| Anxiety Level   | 15.00          |
| Social Activity | 10.00          |

## 6. MODEL VALIDATION

### 6.1 Performance Criteria

Models were evaluated using:

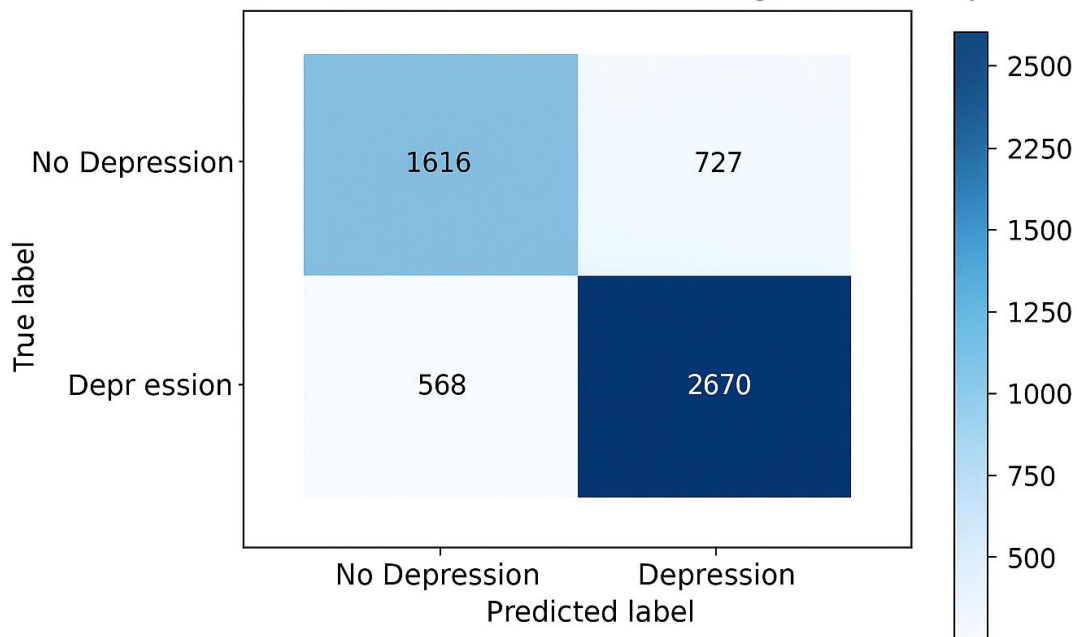
1. Accuracy: Overall prediction correctness.
2. Precision: True positives out of predicted positives.
3. Recall: True positives out of actual positives.
4. F1-Score: Harmonic mean of precision and recall.
5. ROC-AUC: Discrimination ability between depressed and non-depressed classes.

These were chosen to address the class imbalance and ensure robustness in identifying at-risk students.

### 6.2 Testing Results (Random Forest)

| Metric    | Value |
|-----------|-------|
| Accuracy  | 84%   |
| Precision | 0.85  |
| Recall    | 0.81  |
| F1-Score  | 0.83  |
| ROC-AUC   | 0.87  |

Confusion Matrix - Random Forest (Using Cleaned, Imputed Data)



Cross-Validation: 5-fold CV showed <2% standard deviation in accuracy, confirming model consistency across different data splits.

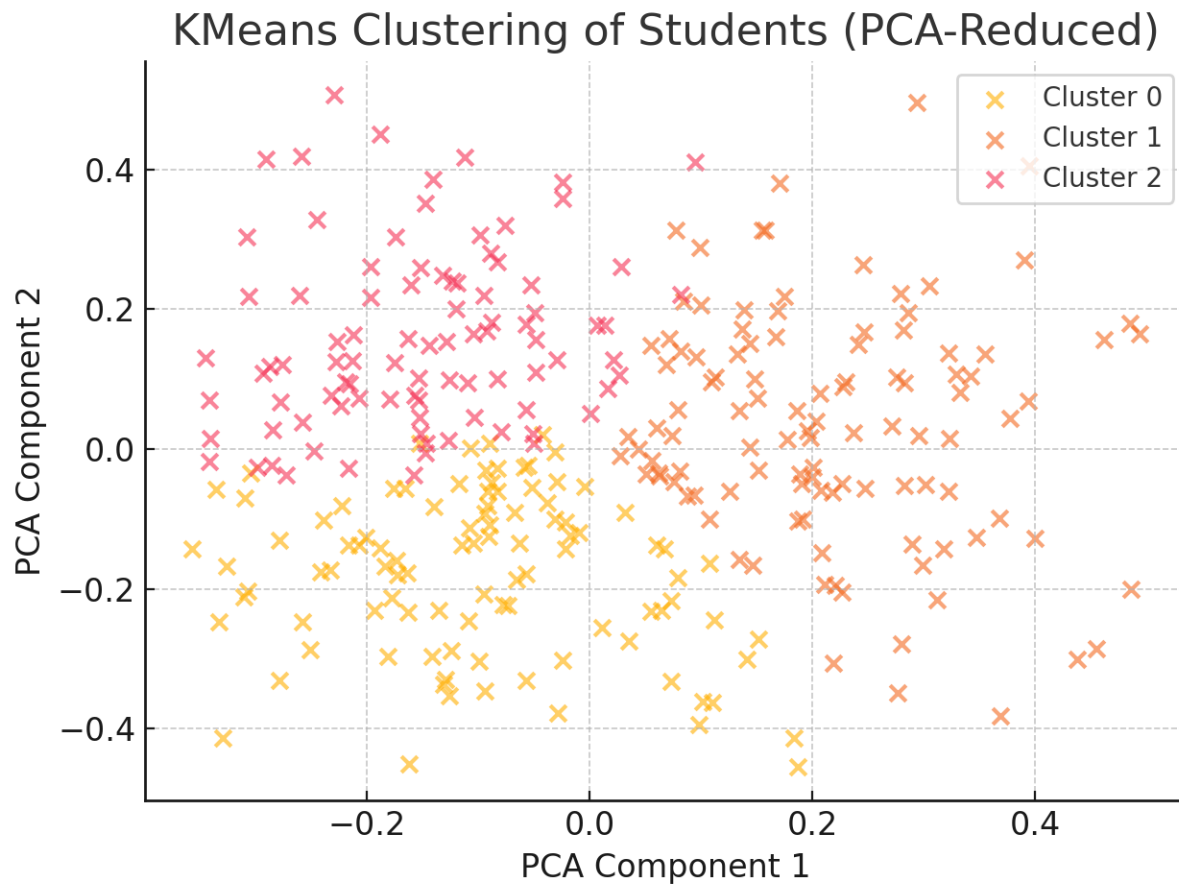
### 6.3 Clustering Analysis

To explore patterns beyond supervised labels, KMeans clustering was applied with k=3:

**Cluster 1:** Low stress, high CGPA — likely well-adjusted students.

**Cluster 2:** Moderate stress, mixed outcomes — transitional group.

**Cluster 3:** High stress, low CGPA, frequent suicidal thoughts — high-risk segment.



This segmentation may inform future personalized interventions or outreach strategies.

## 7. CONCLUSION

This study identified key drivers of student depression using a structured data science pipeline. The Random Forest model achieved strong predictive performance (84% accuracy, 0.87 ROC-AUC), with suicidal thoughts, academic pressure, and financial stress emerging as the most influential factors. While features like sleep duration and dietary habits had moderate impact, they were not sufficient as standalone predictors. Clustering analysis further revealed three distinct student profiles, helping isolate high-risk groups even without labeled outcomes. These insights can guide early intervention strategies and mental health outreach, especially if combined with real-time behavioral data in future work. That said, the dataset's self-reported nature introduces the possibility of bias or underreporting, particularly for sensitive questions. Additionally, the findings are based on a specific population and timeframe, and caution is advised when generalizing. Most importantly, any predictive system based on this approach should be used ethically—to support, not stigmatize, vulnerable students.

### GitHub Repository

[https://github.com/SahilBhaware01/student\\_depression\\_analysis](https://github.com/SahilBhaware01/student_depression_analysis)

## 8. DATA SOURCES

**Dataset Title:** *Student Mental Health Dataset*

**Source:** Kaggle

**Access Link:** <https://www.kaggle.com/datasets/hopesb/student-depression-dataset>

**Format:** CSV file with 27,901 student records and 18 features

**Collected Attributes:** Demographics, academic pressure, work hours, financial stress, CGPA, sleep duration, suicidal thoughts, and depression diagnosis

**Access Info:** Publicly available for download after free Kaggle account registration

## 9. REFERENCES

[1] American College Health Association. *National College Health Assessment*, 2024.

[2] Frontiers in Psychology. “Academic Stress and Mental Health in College Students,” 2019. [Online]. Available: <https://www.frontiersin.org/articles/10.3389/fpsyg.2019.02277/full>

[3] PMC Public Health. “Depression Trends in University Populations,” 2024. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7870388/>

[4] Kaggle. *Student Depression Dataset*. [Online]. Available: <https://www.kaggle.com/datasets/hopesb/student-depression-dataset>

[5] World Health Organization. *Mental Health and Young Adults*, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/adolescent-mental-health>